

lecture9 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a function called `print_in_box(word)` that prints `word` in a box as shown. Use the `print_inBars` function from the previous question to print the middle line of the box. Then ask the user to enter a movie name and print in a box.

```
Enter a movie name: The Matrix
+-----+
|The Matrix|
+-----+
```

Sample Solution

```
def print_inBars(word):
    print(f'|{word}|')

def print_in_box(word):
    bar = '+' + '-' * len(word) + '+'
    print(bar)
    print_inBars(word)
    print(bar)

movie = input("Enter a movie name: ")
print_in_box(movie)
```

Question 2 (q2_sol.py)

Write a program that asks the user to enter an int that is 0 or greater, and a word, and then prints the word in a numbered list as shown. Use a for-loop in your solution. No functions are required, but you can make one if you want.

```
Enter an int: 3
Enter a word: hello!
hello!
hello!
hello!
```

Sample Solution

```
n = int(input("Enter an int: "))
word = input("Enter a word: ")
print()
for i in range(n):
    print(f'{i+1}. {word}')
```

Question 3 (q3_sol.py)

Write a program that asks the user to enter an int (called n) that is 0 or greater. Then, starting at 1, print the ints from 1 to n on the screen in a box as shown. Use the `print_in_box(word)` function from the previous question.

```
Enter an int: 3
```

```
+--+  
|1|  
+--+
```

```
+--+  
|2|  
+--+
```

```
+--+  
|3|  
+--+
```

Sample Solution

```
def print_in_bar(word):  
    print(f'|{word}|')  
  
def print_in_box(word):  
    bar = '+' + '-' * len(word) + '+'  
    print(bar)  
    print_in_bar(word)  
    print(bar)  
  
n = int(input("Enter an int: "))  
print()  
for i in range(1, n+1):  
    print_in_box(str(i))  
    print()
```

Question 4 (q4_sol.py)

Write a program that asks the user to enter an int that is greater than 1, and then prints a countdown from that int to 1, finally printing "Blastoff!".

```
Enter an int: 5
5
4
3
2
1
Blastoff!
```

Sample Solution

```
start = int(input("Enter an int: "))
for i in range(start):
    print(start - i)
print("Blastoff!")
```